

LECTURE SHEET

EXERCISE-I

(Electrolytic conductors, Electrolysis and Faraday Laws)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The reason for increase in electrical conduction of electrolyte with increase in temperature is
 - increase in the number of ions
 - increase in the speed of ions
 - increase in the degree of dissociation of electrolyte
 - all
- Sodium metal in liquid ammonia is
 - an ionic conductor
 - an electronic conductor
 - a mixed conductor
 - non-conductor
- Which of the following is 100% ionised at any dilution?
 - CH_3COOH
 - HCN
 - NaCl
 - NH_4OH
- The degree of dissociation of an electrolyte in aqueous solution depends on
 - Temperature
 - Concentration of the electrolyte
 - Nature of the electrolyte
 - All
- Molten CuCl_2 is electrolysed using platinum electrode. The reaction occurring at anode is
 - $2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$
 - $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-$
 - $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$
 - $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
- During the electrolytic reduction of alumina, the reaction at cathode is
 - $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$
 - $3\text{F}^- \rightarrow 3\text{F} + 3\text{e}^-$
 - $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
 - $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- During the electrolysis of aqueous solution of the following, molarity of the solution increases without changing the chemical composition
 - NaCl
 - HCl
 - CuSO_4
 - H_2SO_4
- (A) : Hydrogen gas always evolved only at cathode during electrolysis
(R) : H^+ ions undergo reduction by gaining electrons
 - Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true and (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - (A) is false but (R) is true
- An aqueous solution containing one mole per litre of each $\text{Cu}(\text{NO}_3)_2$, AgNO_3 , $\text{Hg}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$ is being electrolysed using inert electrodes. The values of standard electrode potential (reduction potential) in volts are
 $\text{Ag} / \text{Ag}^+ = +0.80\text{V}$ $\text{Hg} / \text{Hg}^{2+} = +0.79\text{V}$ $\text{Cu} / \text{Cu}^{2+} = +0.34\text{V}$ $\text{Mg} / \text{Mg}^{2+} = -2.37\text{V}$
 With increasing voltage, the sequence of deposition of metals on cathode will be
 - Ag, Hg, Cu, Mg
 - Mg, Cu, Hg, Ag
 - Ag, Hg, Cu
 - Cu, Hg, Ag
- The electrolysis of an aqueous solution of KNO_3 between platinum electrode gives
 - K at the cathode NO_2 at the anode
 - H_2 at cathode and O_2 at anode
 - H_2 at cathode and NO_2 at anode
 - K at cathode and O_2 at anode

11. Aqueous solution of AgNO_3 is electrolysed using inert electrodes. At the end of electrolysis
- pH of the solution increases
 - pH of the solution decreases
 - pH of the solution remains unchanged
 - pH of the solution becomes 14
12. Find wrong match.
- Very dilute H_2SO_4 by inert electrodes $\Rightarrow \text{H}_2$ at anode
 - Potential is zero Volts $\Rightarrow \text{Pt}, \text{H}_2(1\text{atm})/\text{H}^+(1\text{M})$
 - 50% H_2SO_4 by inert electrodes $\Rightarrow \text{H}_2\text{S}_2\text{O}_8$ at anode
 - $\text{Zn}/\text{Zn}^{2+}(\text{aq})//\text{Cu}^{2+}(\text{aq})/\text{Cu} \Rightarrow \text{Daniel cell}$
13. During the electrolysis of cryolite, aluminium and fluorine are formed in molar ratio
- 1 : 2
 - 2 : 3
 - 1 : 1
 - 1 : 3
14. For the discharge of equal masses of the following ions, the number of electrons required is maximum in the case of
- H^+
 - Cu^{2+}
 - Ag^+
 - Al^{3+}
15. Copper can be deposited from acidified copper sulphate and alkaline cuprous cyanide. If the same current is passed for a definite time :
- The amount of copper deposited from acidic copper sulphate will be higher
 - The amount of copper deposited from alkaline cuprous cyanide will be higher
 - The same amount of copper will be deposited
 - None of the above
16. The no. of electrons involved in the electrolytic deposition of 63.5 g. of Cu from an aq. CuSO_4 is
- 6.0×10^{23}
 - 3.011×10^{23}
 - 12.04×10^{23}
 - 6.02×10^{22}
17. One coulomb of charge passes through solutions of AgNO_3 and CuSO_4 . The ratio of the amounts of silver and copper deposited on platinum electrodes used for electrolysis is
- 108 : 63.5
 - 54 : 31.75
 - 108 : 31.75
 - 215.8 : 31.75
18. The weight in grams of O_2 formed at Pt anode during the electrolysis of an aq. K_2SO_4 solution during the passage of one coulomb of electricity is
- $\frac{16}{96800}$
 - $\frac{8}{96500}$
 - $\frac{32}{96500}$
 - $\frac{64}{96500}$
19. How many coulombs of electricity are required for the reduction of 1 mol of MnO_4^- to Mn^{2+} ?
- 96500 C
 - $1.93 \times 10^5 \text{ C}$
 - $4.83 \times 10^5 \text{ C}$
 - $9.65 \times 10^6 \text{ C}$
20. Electric charge on 1gm ion of N^{3-} is
- $4.8 \times 10^{-19} \text{ C}$
 - $10 \times 1.6 \times 10^{-19} \text{ C}$
 - $1.6 \times 10^{-19} \text{ C}$
 - $2.89 \times 10^5 \text{ C}$
21. Two electrolytic cells, one containing acidified ferrous sulphate and another acidified ferric chloride, are in series. The ratio of masses of Iron deposited at the cathode in the two cells will be
- 3 : 1
 - 2 : 1
 - 1 : 1
 - 3 : 2
22. An ion is reduced to the element when it absorbs 6×10^{20} electrons. The number of equivalents of the ion is
- 0.1
 - 0.01
 - 0.001
 - 0.0001

Numerical Value Type Questions

23. By passing 0.1 F of electricity through fused sodium chloride, the amount of Cl_2 liberated is
24. On electrolysis of a sample of acidified water, 22.4 ml of hydrogen was obtained. The volume of oxygen in ml obtained is
25. An electrolytic cell is constructed for preparing hydrogen. For the average current of 1 ampere in the circuit, the time required for producing 112 ml of H_2 at STP is approximately
26. The charge used for the oxidation of one mole of Mn_3O_4 to MnO_4^{2-} in alkaline medium is ____ F.
27. The electrochemical equivalent of an element is 0.0006735 g/C. Its equivalent weight is ____
28. On passing current through molten KCl , 19.5 g of K is deposited. The amount of Al deposited by the same quantity of electricity when passed through molten AlCl_3 is ____ g.
29. The current strength required to displace 0.1 g. of H_2 in 10 sec is ____ amp.

LEVEL-II (ADVANCED)**Straight Objective Type Questions**

1. A quantity of electrical charge that brings about the deposition of 4.5 g Al from Al^{3+} at the cathode will also produce the following volume (STP) of $\text{H}_2(\text{g})$ from H^+ at the cathode
a) 44.8 L b) 22.4 L c) 11.2 L d) 5.6 L
2. The electrolysis of acetate solution produces ethane according to reaction:
 $2\text{CH}_3\text{COO}^- \rightarrow \text{C}_2\text{H}_6(\text{g}) + 2\text{CO}_2(\text{g}) + 2\text{e}^-$. The current efficiency of the process is 80%. What volume of gases would be produced at 27°C and 740 torr, if the current of 0.5 amp is passed through the solution for 96.45 min? ($V_{\text{C}_2\text{H}_6} + V_{\text{CO}_2}$) = ?
a) 6.0 L b) 0.60 L c) 1.365 L d) 0.91 L
3. A 0.200M KOH solution is electrolysed for 1.5 h using a current of 8.00 A. How many moles of O_2 were produced at the anode?
a) 0.48 b) 0.224 c) 0.112 d) 2.24×10^{-2}
4. A layer of chromium metal 0.25 mm thick is to be plated on an auto bumper with a total area of 0.32m^2 from a solution containing CrO_4^{2-} . What current flow is required for this electroplate if the bumper is to be plated in 60s? The density of chromium metal is 7.20g/cm^3
a) $4.9 \times 10^3\text{A}$ b) $1.78 \times 10^3\text{A}$ c) $5.3 \times 10^4\text{A}$ d) $10.69 \times 10^4\text{A}$
5. Cost of electricity for the production of y L H_2 at STP at cathode is Rs. x, then cost of electricity for the production of y L O_2 gas at STP at anode will be
a) $2x$ b) $4x$ c) $16x$ d) $32x$

More than One correct answer Type Questions

6. Which of the following statements are true ?
a) graphite is an electronic conductor b) Na/liqNH_3 is a mixed conductor
c) 'Si-Ga' allows in a semiconductor d) Aqueous urea solution is not a conductor
7. Which of the following reaction occur during the electrolysis of aq. CuSO_4 using Cu electrodes?
a) $\text{Cu} \rightarrow \text{Cu}^{+2} + 2\text{e}^-$ b) $2\text{SO}_4^{2-} + 2\text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4 + \text{O}_2$
c) $\text{Cu}^{+2} + 2\text{e}^- \rightarrow \text{Cu}$ d) $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
8. In which electrolysis one faraday can produce one mole at cathode or anode?
a) aq. NaCl b) aq. AgNO_3 c) fused NaCl d) aq. CuSO_4

9. Select right statement(s) about electrolysis :
- Electric current is used to drive a non - spontaneous reaction
 - ΔG is positive for chemical process during electrolysis
 - Cations and anions move towards the anode and cathode respectively.
 - Over voltage is generally associated with evolution of O_2 gas in case of aq. solution
10. Coulomb is the quantity of charge defined as :
- One ampere of current passing for 1 sec
 - The charge which deposits 0.001118g of Ag on cathode
 - The charge which deposits electrochemical equivalence of metal
 - 1/2 ampere current for two second
11. An acidic solution of Cu^{2+} salt containing 0.4 g of Cu^{2+} is electrolysed until all the copper is deposited. The electrolysis is continued for seven more minutes with the volume of solution kept at 100 ml and the current at 1.2 amp. Calculate the volume of gases evolved at NTP during the entire electrolysis.
- $H_2 = 48.45\text{ml}$
 - $H_2 = 58.50\text{ml}$
 - $O_2 = 99.8\text{ml}$
 - $O_2 = 100\text{ml}$
12. The charge required for the reduction of 0.4 mol of $K_2Cr_2O_7$ to Cr^{3+} ions is
- 0.6 F
 - $2.4 \times 96500\text{C}$
 - 2.4 F
 - $12.4 \times 96500\text{F}$

Linked Comprehension Type Questions

Passage-I :

Electrolysis is the process in which electrical energy is converted to chemical energy. Amount of substance liberated at electrode is directly proportional to the quantity of charge passed. $w = Eit/96500$

13. The platinum electrodes were immersed in a solution of cupric sulphate ($CuSO_4$) and electric current is passed through the solution. After sometime, it was observed that the colour of copper sulphate disappeared with evolution of a gas at the electrode. The colourless solution contains.
- platinum sulphate
 - copper nitrate
 - copper sulphate
 - sulphuric acid
14. The passage of current liberates H_2 at cathode and Cl_2 at anode. The solution is
- $CuCl_2$ (aq)
 - NaCl in water
 - $HgCl_2$ (aq)
 - $AuCl_3$ in water
15. Calculate the volume of gas liberated at the anode at STP during the electrolysis of an aq. $CuSO_4$ solution by a current of 1 A passed for 16 minutes and 5 seconds.
- 224 mL
 - 56 mL
 - 112 mL
 - 448 mL

Passage-II :

Electrolysis involves electronation and de-electronation at the respective electrodes. Anode of electrolytic cell is the electrode at which de-electronation takes place whereas at cathode electronation is noticed. The products formed at either electrode is given in terms of Faraday's laws of electrolysis i.e. $w = Eit/96500$

16. During electrolysis of $CH_3COONa_{(aq)}$, the mole ratio of gases formed at cathode and anode is
- 1 : 2
 - 2 : 1
 - 3 : 1
 - 1 : 3
17. During electrolysis of $CuSO_{4(aq)}$, the pH of solution becomes
- < 7
 - > 7
 - = 7
 - ≥ 7
18. The volume of octane required to be used for its combustion by the oxygen liberated during electrolysis of $NaNO_3(aq)$ by passing 9.65 ampere current for 1 hr. is
- 322.56 L
 - 32.256 mL
 - 3.22 mL
 - 161.2 mL

Matrix Matching Type Questions19. **Column-I**

- A) Dilute solution of HCl
 B) Dilute solution of NaCl
 C) Concentrate solution of NaCl
 D) Fairly concentrate solution of AgNO_3

Column-II

- p) O_2 evolved at anode
 q) H_2 evolved at cathode
 r) Cl_2 evolved at anode
 s) Ag deposition at cathode

20. **Column-I**

- A) 1 mole of MnO_4^- to Mn^{2+}
 B) 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+}
 C) 1 mole of Sn^{4+} to Sn^{2+}
 D) 1 mole of Al^{3+} to Al

Column-II

- p) 193000 C
 q) 289500 C
 r) 482500 C
 s) 579000 C

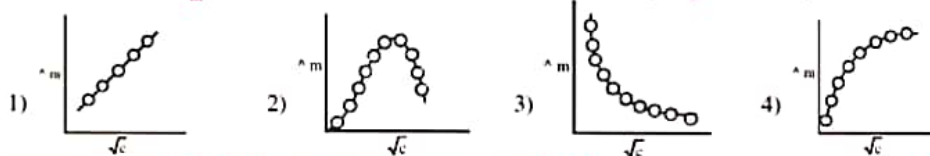
Integer Type Questions

21. A current of 2 ampere passing for 5 hr through a molten tin salt deposits 22.2 g of tin. Find the oxidation number of tin (atomic weight of tin is 118.7) in the salt
22. What current is to be passed for 0.25 sec for deposition of certain weight of metal which is equal to its electrochemical equivalent ?
23. How many faradays are required for reduction of 1 mol $\text{C}_6\text{H}_5\text{NO}_2$ into $\text{C}_6\text{H}_5\text{NH}_2$
24. What is the equivalent weight of O_2 in the following reaction, $\text{H}_2\text{O} + \frac{1}{2} \text{O}_2 + 2\text{e}^- \rightarrow 2\text{OH}^-$?
25. During the electrolysis of a concentrated brine solution. Calculate the moles of chlorine gas produced by the passage of $4F$ electricity.
26. Total charge required for the oxidation of two moles Mn_3O_4 into MnO_4^{2-} in presence of alkaline medium is 2.0×10^x ; x is
27. In the oxidation of ferric oxalate $[\text{Fe}_2(\text{C}_2\text{O}_4)_3]$ to carbondioxide, if $18F$ of electricity is required how many moles of ferric oxalate is oxidized.
28. The electricity in term of faraday is required to produce 20g of Ca from molten CaCl_2 is:

EXERCISE-II*(Electrolytic Conductance)***LEVEL-I (MAIN)**Straight Objective Type Questions

1. The ionic conductance of following cations in a given conc. aq. solution is in the order
 1) $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+$ 2) $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+$ 3) $\text{Li}^+ < \text{Na}^+ > \text{K}^+ > \text{Rb}^+$ 4) $\text{Li}^+ = \text{Na}^+ < \text{K}^+ < \text{Rb}^+$
2. (A) : The conductivity of an aqueous solution of NaCl is greater than that of pure solvent.
 (R) : Conductivity is independent of the number of ions in solution.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
3. The highest electrical conductivity among the following aqueous solutions is of
 1) 0.1 M acetic acid 2) 0.1 M chloroacetic acid
 3) 0.1 M fluoroacetic acid 4) 0.1 M difluoroacetic acid

4. Conductance is directly proportional to area and the concentration of solution in it and is inversely proportional to the length of the vessel, then the unit of the constant of proportionality is
 1) $S.m\ mol^{-1}$ 2) $S.m^2\ mol^{-1}$ 3) $S^{-2}.m^2\ mol$ 4) $S^2.m^2\ mol^{-2}$
5. If the specific conductance and conductance of a solution are same, then its cell constant is equal to
 1) 1 2) 0 3) 10 4) 100
6. (A) : The conductivity of 0.1M solutions of different electrolytes is same.
 (R) : The conductivity depends on the size of the ions.
7. Specific conductivity of a solution
 1) increases with dilution 2) decreases with dilution
 3) remains unchanged with dilution 4) depends on mass of electrolyte
8. A graph is drawn between the λ_{eq} values and concentration of an electrolyte. Which of the following electrolyte will correspond to the graph given ?
 1) KCl
 2) $BaCl_2$
 3) H_2SO_4
 4) CH_3COOH
9. For which case ' λ ' values v/s $\sqrt{C}$ shows a straight line
 1) KCl 2) $HCOOH$ 3) CH_3NH_2 4) CH_3COOH
10. The expression showing the relationship between equivalent conductivity and molar conductivity is
 1) $\lambda_m = Z \times \lambda_{eq}$ 2) $\lambda_{eq} = Z \times \lambda_m$ 3) $\lambda_m = \frac{\lambda_{eq}}{Z}$ 4) $\lambda_m = \lambda_{eq}^2$
11. At infinite dilution stage, the solution of CH_3COOH in water does not contain:
 1) H^+ ion 2) CH_3COO^- ion 3) CH_3COOH molecule 4) H_2O
12. Which of the following solutions of NaCl has the higher specific conductance ?
 1) 0.001N 2) 0.01N 3) 0.1 N 4) 1 N
13. The equivalent conductances of two strong electrolytes at infinite dilution in H_2O (where ions move freely through a solution) at $25^\circ C$ are given below $\Lambda_{CH_3COONa}^\circ = 91.0\ S\ cm^2 / equiv$; $\Lambda_{HCl}^\circ = 426.2\ S\ cm^2 / equiv$. What additional information / quantity one needs to calculate Λ° of an aqueous solution of acetic acid ?
 1) Λ° of CH_3COOK 2) Λ° of H^+ 3) Λ° of $ClCH_2COOH$ 4) Λ° of NaCl
14. For a dilute solution of a strong electrolyte, which of the following facts is correct?
 1) The graph b/w λ_m and C is linear
 2) The graph b/w $\log \lambda_m$ and C is linear
 3) The graph b/w λ_m and $\sqrt{C}$ is not linear
 4) The graph b/w λ_m and $\sqrt{C}$ has a negative slope.
15. The variation of Λ_m of acetic acid with concentration is correctly represented by



16. CH_3COOH is titrated in the NaOH solution then which of the following statements is correct?
- Conductance increases upto equivalence point, then it decreases.
 - Conductance decreases upto equivalence point, then it increases.
 - First conductance increases slowly upto equivalence point and then increases rapidly.
 - Before the equivalent point, there exists basic buffer solution.
17. $\text{AgNO}_{3(aq)}$ was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance Λ versus the volume of AgNO_3 is



18. Which one is correct about conductivity water ?
- The water whose own conductance is very small
 - The water obtained after 7-8 times distillation
 - Kohlrausch prepared the conductivity water for the first time
 - All of these

Numerical Value Type Questions

19. The distance b/w two electrodes of a cell is 2.5cm and area of electrode is 5cm^2 the cell constant is
20. For an electrolytic solution of $0.05\text{ mole litre}^{-1}$, the conductivity has been found to be 0.0110 Scm^{-1} . The molar conductivity (in $\text{Scm}^2\text{mole}^{-1}$) is
21. The limiting molar conductivities (Λ^0) for NaCl , KBr and KCl are 126, 152 and $150\text{ S.cm}^2\text{mol}^{-1}$ respectively. Then Λ^0 for NaBr is
22. Molar ionic conductivities of a bivalent electrolyte are 57 and 73. The Λ_m of the solution will be
23. The conductivity of 0.001 M acetic acid is $5 \times 10^{-5}\text{ S cm}^{-1}$ and Λ^0 is $390.5\text{ S cm}^2\text{mol}^{-1}$ then the calculated value of dissociation constant of acetic acid would be $____ \times 10^{-6}$
24. The equivalent conductivity of $0.1\text{ N CH}_3\text{COOH}$ at 25°C is 80 and at infinite dilution is $400\text{ohm cm}^2\text{eq}^{-1}$. The degree of dissociation of CH_3COOH
25. The equivalent conductance at infinite dilution of NaCl , HCl and CH_3COONa at 298K are 126.0, 426.0 and $91.0\text{ ohm}^{-1}\text{cm}^2$ respectively. The value of equivalent conductance of acetic acid at infinite dilution at the same temperature is
26. Given the following molar conductivities at 25°C : HCl , $426\text{ }\Omega^{-1}\text{cm}^2\text{mol}^{-1}$; NaCl , $126\text{ }\Omega^{-1}\text{cm}^2\text{mol}^{-1}$; NaC (sodium crotonate), $83\text{ }\Omega^{-1}\text{cm}^2\text{mol}^{-1}$, $x \times 10^{-5}$ is the ionization constant of crotonic acid, if the conductivity of a 0.001 M crotonic acid solution is $3.83 \times 10^{-5}\text{ }\Omega^{-1}\text{cm}^{-1}$, $x =$
27. Equivalent conductance of saturated BaSO_4 is $400\text{ohm}^{-1}\text{cm}^2\text{equiv}^{-1}$ and specific conductance is $8 \times 10^{-5}\text{ohm}^{-1}\text{cm}^{-1}$. Hence, solubility of BaSO_4 is $____ \times 10^{-4}\text{M}$.

LEVEL-II (ADVANCED)

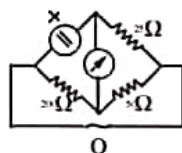
Straight Objective Type Questions

1. The specific conductivity of dichloroacetic acid at dilution of 8 litres is 0.0238 C.G.S. unit, λ_0 for this acid is 385 C.G.S. unit, calculate the degree of ionisation at this dilution.
- a) 50% b) 45% c) 49.45% d) 36.02%

2. $\lambda_m^0(\text{AgNO}_3)$, $\lambda_m^0(\text{HCl})$ and $\lambda_m^0(\text{HNO}_3)$ are a , b and c $\text{S cm}^2 \text{mol}^{-1}$ respectively. If the conductivity of a saturated solution is Z S cm^{-1} . Then the solubility of AgCl is
- a) $\frac{a+b+c}{1000MZ}$ b) $\frac{(a+b+c)1000}{Z}$ c) $\frac{1000Z}{a+b-c}$ d) $\frac{Z(a+b-c)}{1000M}$
3. The specific conductance of a saturated solution of a salt MX is $2.8 \times 10^{-7} \Omega^{-1} \text{cm}^{-1}$ and the equivalent conductances at infinite dilution of M^{+1} and X^{-1} ions are 68 and $72 \Omega^{-1} \text{cm}^2/\text{equivalent}$ respectively. Hence, the K_{sp} of MX is found to be
- a) 2×10^{-6} b) 4×10^{-12} c) 10^{-12} d) 4×10^{-6}
4. The resistance of 0.01 N solution of an electrolyte is 210 ohm at 298K , its conductance is :
- a) 4.76×10^{-3} mho b) 4.76 mho c) 210 mho d) none of these
5. If the electrolyte used is $\text{Ba}(\text{NO}_3)_2$, then its molar conductance to equivalent conductance ratio is
- a) 1 b) 2 c) 4 d) 3

More than One correct answer Type Questions

6. The molar conductivities of H^+ and HCOO^- ions at infinite dilution are 34.6 and $5.4 \text{ Sm}^2 \text{mol}^{-1}$ respectively. The molar conductivity of 0.025M HCOOH is $4.0 \text{ Sm}^2 \text{mol}^{-1}$. Then correct one for formic acid is
- a) $\text{pK}_a = 3.6$ b) $\text{K}_a = 2 \times 10^{-3}$ c) $\text{pK}_a = 4.8$ d) $\text{K}_a = 2.5 \times 10^{-4}$
7. Limiting equivalent conductivities of different ions are given below.
- | Ion | K^+ | Na^+ | SO_4^{2-} |
|--|--------------|---------------|--------------------|
| $\lambda_{eq}^0 / \text{Scm}^2 \text{eq}^{-1}$ | 73 | 81 | 58.8 |
- The limiting equivalent conductivity of the salt NaKSO_4 is ____ $\text{Scm}^2 \text{eq}^{-1}$
- a) 239.8 b) 201.4 c) 135.8 d) 141.6
8. A conductivity cell (X) employs two electrodes having surface area of 10 cm^2 and separated by a distance of 10 cm . It is filled with 0.01M solution of an electrolyte and connected in the wheatstone bridge. The balanced bridge is shown in the diagram. What is the molar conductivity of the electrolyte?



- a) $10^3 \text{ Scm}^2 \text{mol}^{-1}$ b) $250 \text{ Scm}^2 \text{mol}^{-1}$ c) $100 \text{ Scm}^2 \text{mol}^{-1}$ d) $500 \text{ Scm}^2 \text{mol}^{-1}$

Linked Comprehension Type Questions

Passage-I:

Molar conductivity, $\Lambda_m = K/C$ where K is conductivity (S cm^{-1}), ($\Lambda_m = \text{Scm}^2 \text{mol}^{-1}$) and C is molar concentration (in mole $/\text{cm}^3$). The molar conductivity of 0.04 M solution of MgCl_2 is $200 \text{ Scm}^2 \text{mol}^{-1}$ at 298 K . A cell with electrodes that are 2.0 cm^2 in surface area and 0.50 cm apart is filled with MgCl_2 solution.

9. Conductivity of MgCl_2 solution is :
- a) $8 \times 10^{-3} \text{ S cm}^{-1}$ b) 32 S cm^{-1} c) 0.032 S cm^{-1} d) None of these

10. How much current will flow when the potential difference between the two electrodes is 5.0V?
 a) 156.25 A b) 0.16 A c) 160 A d) None of these
11. The cell constant is
 a) 0.25cm^{-1} b) 0.50cm^2 c) 0.025cm d) 0.35cm^2

Passage-II :

Specific conductance or conductivity of the electrolyte solution is given by the following relation: $K = G \cdot l / A$ where, $G = 1/R$ is the conductance and l/A is the cell constant, Molar conductance (Λ_m) and equivalent conductance (Λ_e) of an electrolyte solution are calculated using the following similar relations: $\Lambda_m = K \cdot 1000 / M$; $\Lambda_e = K \cdot 1000 / N$ where, M and N are the molarity and normality of the solution respectively. Molar conductance of strong electrolyte depends on concentration: $\Lambda_m = \Lambda_m^0 - b\sqrt{C}$ where, Λ_m^0 = molar conductance at infinite dilution; C = concentration of the solution b = constant. $\alpha = \Lambda_m / \Lambda_m^0 = \Lambda_e / \Lambda_e^0$

12. Which of the following decreases on dilution of electrolytic solution?
 a) Λ_{eq} b) Λ_m c) K d) G
13. For which of the following electrolytic solution Λ_m and Λ_e are equal ?
 a) BaCl_2 b) KCl c) $\text{Al}_2(\text{SO}_4)_3$ d) CaCl_2
14. Which of the following equality holds good for the strong electrolytes?
 a) $\Lambda = \Lambda^0$ as $C \rightarrow 1$ b) $\Lambda = \Lambda^0$ as $C \rightarrow 0$ c) $\Lambda = \Lambda^0$ as $C \rightarrow \infty$ d) $\Lambda = \Lambda^0$ as $C \rightarrow \sqrt{b}$

Matrix Matching Type Questions

15. **Column-I (In fused state)** **Column-II (Conductivity in $\text{Ohm}^{-1}\text{cm}^{-1}$)**
 A) H^+ p) 350
 B) Na^+ q) 50
 C) Li^+ r) 39
 D) Cs^+ s) 77
16. **Column-I (Term)** **Column-II (Relation)**
 A) α (Degree of ionization) p) λ^+ / λ
 B) t_+ (Transport number) q) $\Lambda_m^c / \Lambda_m^0$
 C) Fraction of a mole undergoing ionization r) $U^+ / U^+ + U^-$
 D) Transport number of Na^+ ion in NaCl s) $\Lambda_e^c / \Lambda_e^0$

Integer Type Questions

17. The ratio of (Λ_m / Λ_e) for $\text{Ca}_3(\text{PO}_4)_2$ will be equal to _____
18. The resistance of 0.1 N solution of formic acid is 200 ohm and cell constant is 2.0cm^{-1} . The equivalent conductivity ($\text{in Sm}^2\text{eq}^{-1}$) of 0.1 N formic acid is 1.0×10^x ; x is _____
19. The conductance of a salt solution (AB) measured by two parallel electrodes of area 100cm^2 separated by 10 cm was found to be $0.0001\Omega^{-1}$. If volume enclosed between two electrode contains 0.1 mole of salt, and the molar conductivity ($\text{Sm}^2\text{mol}^{-1}$) of salt at same concentration is 1.0×10^{-x} ; x is _____
20. Molar conductivities at infinite dilution of KCl , HCl and CH_3COOK are 0.013, 0.038 and $0.009\text{ Sm}^2\text{mol}^{-1}$ respectively at 291K. If conductivity of 0.001M CH_3COOH is $2.72 \times 10^{-3}\text{ Sm}^{-1}$ then find % degree of ionization of CH_3COOH .

21. For a saturated solution of AgCl at 25°C, specific conductance is $3.41 \times 10^{-6} \text{ ohm}^{-1}\text{cm}^{-1}$. If $\lambda_{\text{AgCl}}^{\infty} = 138.3 \text{ ohm}^{-1}\text{cm}^2\text{eq}^{-1}$, the K_{sp} of AgCl is $0.17 \times 10^{-3} \text{ M}^2$. The value of 'x' is
22. The equivalent conductance of 1.0M CH_3COOH is $12 \Omega^{-1}\text{cm}^2\text{eq}^{-1}$ and that at infinite dilution is $240 \Omega^{-1}\text{cm}^2\text{eq}^{-1}$. The % ionization of CH_3COOH is :

EXERCISE-III

(Galvanic cell & Nernst Equation)

LEVEL-I (MAIN)

Straight Objective Type Questions

- In a Galvanic cell, the electrons flow from
 - anode to cathode through the solution
 - cathode to anode through the solution
 - anode to cathode through the external circuit
 - cathode to anode through the external circuit
- Which of the following statements is wrong about galvanic cells
 - cathode is the positive electrode
 - cathode is the negative electrode
 - e^- flow from anode to cathode in the external circuit
 - reduction occurs at cathode
- Which of the following statements is correct w.r.t. both electrolytic cell and Galvanic cell
 - in both cells, anode is shown by +ve sign
 - in both cells, cathode is shown by -ve sign
 - in both cells, reduction reaction takes place at the cathode
 - in both cells, oxidation reaction takes place at the cathode
- Saturated solution of KNO_3 is used to make salt bridge because
 - velocity of K^+ is greater than that of NO_3^-
 - velocity of NO_3^- is greater than that of K^+
 - velocities of K^+ and NO_3^- are $\approx$ the same
 - KNO_3 is highly soluble in water
- (A) : A salt bridge allows the flow of current by completing the electrical circuit.
(R) : A salt bridge maintains the electrical neutrality of the two half cells.
 - Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true and (R) is not the correct explanation of (A)
 - (A) is true but (R) is false
 - (A) is false but (R) is true
- Which of the following is correct?
 - Zinc acts as cathode in Daniel cell
 - In a Li-Zn couple, zinc acts as anode
 - Copper displaces iron from its salt solution
 - Zinc displaces tin from its salt solution
- Which of the following statements is true for the electrochemical Daniel cell ?
 - e^- flow from Cu electrode to Zn electrode
 - Current flows from Zn electrode to Cu electrode
 - Cations move towards copper electrode
 - Cations move toward zinc electrode
- The cell reaction of the galvanic cell, $\text{Cu(s)} / \text{Cu}^{2+}(\text{aq}) // \text{Hg}_2^{2+}(\text{aq}) // \text{Hg(l)}$ is
 - $\text{Hg} + \text{Cu}^{2+} \rightarrow \text{Hg}_2^{2+} + \text{Cu}$
 - $\text{Hg} + \text{Cu}^{2+} \rightarrow \text{Hg}^+ + \text{Cu}^+$
 - $\text{Hg} + \text{Cu}^+ \rightarrow \text{CuHg}$
 - $\text{Cu} + \text{Hg}_2^{2+} \rightarrow \text{Cu}^{2+} + \text{Hg}$

9. (A) : The Daniel cell becomes dead after some time.
 (R) : Oxidation potential of zinc anode increases and that of copper cathode decreases.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false
 4) (A) is false but (R) is true
10. The cell for which the cell reaction is $\text{H}_2 + \text{Cu}^{2+} \rightarrow 2\text{H}^+ + \text{Cu}$ is represented as
 1) $\text{Cu}/\text{Cu}^{2+} // \text{H}^+ / \text{H}_2$ 2) $\text{H}_2(\text{g}) / \text{H}^+ // \text{Cu}^{2+} / \text{Cu}$
 3) $\text{Pt}, \text{H}_2(1 \text{ atm}), \text{H}^+ // \text{Cu}^{2+} / \text{Cu}$ 4) $\text{Pt}, \text{H}_2 / \text{H}^+(\text{aq}) (1 \text{ atm}) // \text{Cu}^{2+} / \text{Cu}$
11. Which metal will dissolve if the cell works $\text{Cu} | \text{Cu}^{2+} || \text{Ag}^+ | \text{Ag}$
 1) Cu 2) Ag 3) Both (1) and (2) 4) None of these
12. Which is a correct cell reaction ?
 1) $\text{Zn} + 2\text{Ag}^+ \rightarrow \text{Zn}^{2+} + 2\text{Ag}$ 2) $2\text{Ag} + \text{Zn}^{2+} \rightarrow 2\text{Ag}^+ + \text{Zn}$
 3) Both 4) None
13. The reaction, $1/2\text{H}_{2(\text{g})} + \text{AgCl}_{(\text{s})} \rightarrow \text{H}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} + \text{Ag}_{(\text{s})}$ occurs in the galvanic cell :
 1) $\text{Ag} | \text{AgCl}_{(\text{s})} | \text{KCl}_{(\text{Soln})} | \text{AgNO}_3_{(\text{Soln})} | \text{Ag}$ 2) $\text{Pt}, \text{H}_{2(\text{g})} | \text{HCl}_{(\text{Soln})} | \text{AgNO}_3_{(\text{Soln})} | \text{Ag}$
 3) $\text{Pt}, \text{H}_{2(\text{g})} | \text{HCl}_{(\text{Soln})} | \text{AgCl}_{(\text{s})} | \text{Ag}$ 4) $\text{Pt}, \text{H}_{2(\text{g})} | \text{KCl}_{(\text{Soln})} | \text{AgCl}_{(\text{s})} | \text{Ag}$
14. For spontaneity of a cell, which is correct ?
 1) $\Delta G = 0, \Delta E = 0$ 2) $\Delta G = -Ve, \Delta E = 0$ 3) $\Delta G = +Ve, \Delta E = 0$ 4) $\Delta G = -Ve$
15. If the cell reaction is spontaneous
 1) E^0 is -ve 2) ΔG is positive 3) E^0 is +ve 4) ΔG and E^0 are positive
16. (A) : If E^0 of $\text{Cu}^{2+}/\text{Cu} = +0.34\text{V}$ and E^0 of $\text{Ag}^+/\text{Ag} = +0.80\text{V}$ then galvanic cell constructed from these is $\text{Ag}/\text{Ag}^+//\text{Cu}^{2+}/\text{Cu}$
 (R) : In any galvanic cell the reaction that takes place is a redox reaction
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false
 4) (A) is false but (R) is true
17. A galvanic cell is composed of two hydrogen electrodes, one of which is a standard one. In which of the following solutions should the other electrode be immersed to get maximum EMF?
 1) 0.1 M HCl 2) 0.1 M CH_3COOH 3) 0.1 M H_3PO_4 4) 0.1 M H_2SO_4
18. For the cell $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$, if the conc. of Zn^{2+} and Cu^{2+} ions is doubled, the EMF of the cell:
 1) doubles 2) reduces to half 3) remains same 4) becomes zero
19. For which cell EMF is independent of the concentration of electrolytes used?
 1) $\text{Fe} | \text{FeO}_{(\text{s})} | \text{KOH}_{(\text{aq})} | \text{NiO}_{(\text{s})} | \text{Ni}_2\text{O}_{3(\text{s})} | \text{Ni}$ 2) $\text{Pt}(\text{H}_2) | \text{HCl} | \text{Pt}(\text{Cl}_2)$
 3) $\text{Zn} | \text{Zn}(\text{NO}_3)_2 | \text{CuSO}_4 | \text{Cu}$ 4) $\text{Hg}, \text{HgCl}_2 | \text{KCl} | \text{AgNO}_3 | \text{Ag}$

20. $\text{Pb}_{(s)} \mid \underset{\text{saturated solution}}{\text{PbSO}_4} \mid \mid \underset{\text{saturated solution}}{\text{PbI}_2} \mid \text{Pb}_{(s)}$. Which of the following expressions represent the EMF of the above cell at 25°C ?

$$1) E = \frac{0.0592}{2} \log \frac{[a_{\text{Pb}^{2+}}]_{\text{RHS}}}{[a_{\text{Pb}^{2+}}]_{\text{LHS}}}$$

$$2) E = \frac{0.0592}{2} \log \frac{[a_{\text{Pb}^{2+}}]_{\text{LHS}}}{[a_{\text{Pb}^{2+}}]_{\text{RHS}}}$$

$$3) E = \frac{0.0592}{2} \log [K_{\text{sp}(\text{PbI}_2)}]^{1/2}$$

$$4) E = \frac{0.0592}{2} \log \frac{K_{\text{sp}(\text{PbI}_2)}}{K_{\text{sp}(\text{PbSO}_4)}}$$

21. Given, $\text{Cu}^{2+} + e^- \rightarrow \text{Cu}^+ E^0 = 0.15 \text{ volt}$; $\text{Cu}^+ + e^- \rightarrow \text{Cu} E^0 = 0.5 \text{ volt}$. E for $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$
- 1) 0.65 V 2) 0.325 V 3) 0.45 V 4) 1.2 V

Numerical Value Type Questions

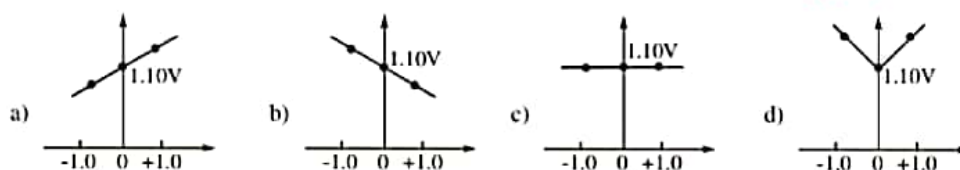
22. If the standard electrode potential of Cu^{2+}/Cu electrode is 0.34 V, what is the electrode potential at 0.01 M concentration of Cu^{2+} ? ($T = 298^\circ\text{K}$)
23. The potential of hydrogen electrode is -118 mV . The H^+ concentration of the solution is
24. The standard potentials (E^0) for the half reactions are as $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$, $E^0 = +0.76 \text{ V}$
 $\text{Fe} \rightarrow \text{Fe}^{2+} + 2e^-$, $E^0 = +0.41 \text{ V}$. The emf for the cell reaction $\text{Fe}^{2+} + \text{Zn} \rightarrow \text{Zn}^{2+} + \text{Fe}$ is
25. Consider the following E^0 values $E^0_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +0.77 \text{ V}$; $E^0_{\text{Sn}^{2+}/\text{Sn}} = -0.14 \text{ V}$. Under standard conditions the potential for the reaction $\text{Sn}_{(s)} + 2\text{Fe}^{3+}_{(\text{aq})} \rightarrow 2\text{Fe}^{2+}_{(\text{aq})} + \text{Sn}^{2+}_{(\text{aq})}$ is
26. E.M.F of the cell reaction, $2\text{Ag}^+ + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}^{2+}$ is 0.46 V. If $E^0_{\text{Cu}^{2+}/\text{Cu}}$ is +0.34 V, $E^0_{\text{Ag}^+/\text{Ag}}$ is
27. If the SRP of nickel and chlorine electrodes are -0.25 V and $+1.36 \text{ V}$ respectively. The EMF of the cell $\text{Ni}/\text{Ni}^{2+} (0.01 \text{ M}) \parallel \text{Cl}^- (0.01 \text{ M})/\text{Cl}_2$, Pt is ____ V
28. The standard electrode potential of the two half cells are given below. $\text{Ni}^{2+} + 2e^- \rightarrow \text{Ni}$; $E^0 = -0.25 \text{ V}$;
 $\text{Zn}^{2+} + 2e^- \rightarrow \text{Zn}$; $E^0 = -0.77 \text{ V}$. The voltage of cell formed by combining the two half cells would be

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. For the cell $\text{Zn} \mid \text{Zn}^{2+} (C_1) \parallel \text{Zn}^{2+} (C_2) \mid \text{Zn}$. ΔG is negative if :
- a) $C_1 = C_2$ b) $C_1 > C_2$ c) $C_2 > C_1$ d) None of these
2. The Nernst eqn., $E = E^0 - \frac{RT}{nF} \log Q$, the equilibrium constant K_c will be equal to Q when :
- a) $E = E^0$ b) $\frac{RT}{nF} = 1$ c) $E = \text{zero}$ d) $E^0 = 1$
3. The standard emf of the cell, $\text{Pt}, \text{H}_2 / \text{H}^+ (C_1) \parallel \text{H}^+ (C_2) / \text{H}_2, \text{Pt}$, is :
- a) $\pm 0.00 \text{ V}$ b) $+1.36 \text{ V}$ c) -1.36 V d) insufficient data
4. Which statement is true about a spontaneous cell reaction in galvanic cell?
- a) $E^0_{\text{cell}} > 0$, $\Delta G^0 < 0$, $Q < K_c$ b) $E^0_{\text{cell}} > 0$, $\Delta G^0 > 0$, $Q < K_c$
c) $E^0_{\text{cell}} > 0$, $\Delta G^0 > 0$, $Q > K_c$ d) $E^0_{\text{cell}} < 0$, $\Delta G^0 < 0$, $Q < K_c$

5. Which graph correctly correlates E_{cell} as a function of concentrations for the cell (for different values of M and M'): $\text{Zn}_{(s)} + \text{Cu}_{(M')}^{2+} \rightarrow \text{Zn}_{(M)}^{2+} + \text{Cu}_{(s)}$; $E_{\text{cell}}^0 = 1.10\text{V}$; X-axis: $\log_{10} \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$; Y-axis: E_{cell}



6. The standard reduction potential of Cu^{2+}/Cu and Cu^+/Cu are 0.337 and 0.153V respectively. The standard electrode potential of $\text{Cu}^{2+}/\text{Cu}^+$ half cell will be
 a) 0.184 V b) 0.827 V c) 0.521 V d) 0.490 V
7. What is reaction quotient, Q, for the cell $\text{Ni}_{(s)} | \text{Ni}^{2+}(0.190\text{M}) || \text{Cl}^- (0.40\text{M}) | \text{Cl}_{2(g)} | \text{Pt}_{(s)} (1.0\text{atm})$
 a) 3.1×10^{-1} b) 1.3×10^{-1} c) 8.0×10^{-2} d) 3.0×10^{-2}
8. Consider the following standard reduction potentials. $\text{Fe}^{3+}(\text{aq}) + e^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$, $E^0 = 0.77\text{V}$, $\text{H}_2\text{O}_2(\text{aq}) + 2e^- \rightleftharpoons 2\text{OH}^-(\text{aq})$, $E^0 = 0.88\text{V}$. For the voltaic cell reaction below, calculate the Fe^{2+} concentration (in M) that would be needed to produce a cell potential equal to 0.16V at 25°C when $[\text{OH}^-] = 0.1\text{M}$, $[\text{Fe}^{3+}] = 0.5\text{M}$ and $[\text{H}_2\text{O}_2] = 0.35\text{M}$: $2\text{Fe}^{2+}(\text{aq}) + \text{H}_2\text{O}_2(\text{aq}) \rightleftharpoons 2\text{Fe}^{3+}(\text{aq}) + 2\text{OH}^-(\text{aq})$
 a) 0.3M b) 0.6M c) 0.41M d) 0.35M
9. $\Delta G = \Delta H - T\Delta S$ and $\Delta G = \Delta H + T \left[\frac{d(\Delta G)}{dT} \right]$ then $\left(\frac{dE_{\text{cell}}}{dT} \right)$ is
 a) $\frac{\Delta S}{nF}$ b) $\frac{nE}{\Delta S}$ c) $-nFE_{\text{cell}}$ d) $+nFE_{\text{cell}}$

More than One correct answer Type Questions

10. Which of the following facts regarding the movement of anions in the solution is incorrect?
 a) Towards cathode in an electrolyte cell and towards anode in a galvanic cell.
 b) Towards anode in an electrolytic cell and towards cathode in a galvanic cell.
 c) Towards cathode in both type cells. d) Towards anode in both type cells.
11. E^0 for $\text{Cl}_2 + 2e^- \rightleftharpoons 2\text{Cl}^-$ is E_1^0 then E_2^0 for $\frac{1}{2}\text{Cl}_2 + 1e^- \rightleftharpoons \text{Cl}^-$ is (If Pt , E^0 of $\frac{1}{2}\text{Cl}_2 | \text{Cl}^- = +1.36\text{V}$)
 a) $E_1^0 = +1.36\text{V}$ b) $E_2^0 = +1.36\text{V}$ c) $E_1^0 = +2.72\text{V}$ d) $E_2^0 = +0.34\text{V}$
12. Which of the following statement is correct?
 If $E_{\text{Cu}^{2+}|\text{Cu}}^0 = 0.34\text{V}$ and $E_{\text{Sn}^{2+}|\text{Sn}}^0 = -0.136\text{V}$, $E_{\text{H}^+|\text{H}_2}^0 = -0.0\text{V}$
 a) Cu^{2+} ions can be reduced by $\text{H}_2(\text{g})$ b) Cu can be oxidized by H^+
 c) Sn^{2+} ions can be reduced by H_2 d) Sn can be oxidized by Cu^{2+}
13. The oxidation potential of hydrogen half-cell will be negative if :
 a) $p(\text{H}_2) = 1\text{atm}$ and $[\text{H}^+] = 1\text{M}$ b) $p(\text{H}_2) = 1\text{atm}$ and $[\text{H}^+] = 2\text{M}$
 c) $p(\text{H}_2) = 0.2\text{atm}$ and $[\text{H}^+] = 1\text{M}$ d) $p(\text{H}_2) = 0.2\text{atm}$ and $[\text{H}^+] = 0.2\text{M}$
14. The cell reaction for the given cell is : $\text{Pt}(\text{H}_2) | \text{pH} = (X) || \text{pH} = 3 | (\text{H}_2), \text{Pt}$, it may be
 $P_1 = 1\text{atm}$ $P_2 = 1\text{atm}$
 a) Spontaneous b) Non-spontaneous
 c) In equilibrium d) It is a half cell

15. Which represents a concentration cell?
- a) $\text{PtH}_2 | \text{HCl} | \text{HCl} | \text{PtH}_2$
 $\text{c}_1 \quad \text{c}_2$
- b) $\text{PtH}_2 | \text{HCl} | \text{Cl}_2 | \text{Pt}$
 c_1
- c) $\text{Zn} | \text{Zn}^{2+}(\text{C}_1) || \text{Zn}^{2+}(\text{C}_2) | \text{Zn}$
- d) $\text{Pt, Cl}_2 | \text{Cl}^{(-)}(\text{C}_1) || \text{Cl}^{(-)}(\text{C}_2) | \text{Cl}_2, \text{Pt}$
16. For the cell at 298 K $\text{Ag}_{(\text{s})} | \text{AgCl}_{(\text{s})} | \text{KCl}_{(\text{aq})} | \text{AgNO}_3_{(\text{aq})} | \text{Ag}_{(\text{s})}$ Which of the following is wrong
- a) The EMF of the cell is zero when $[\text{Ag}^+]_{\text{anodic}} = (\text{Ag}^+)_{\text{cathodic}}$
- b) The amount of $\text{AgCl}_{(\text{s})}$ ppt. in anodic compartment will decrease with the working of the cell
- c) The concentration of $[\text{Ag}^+]$ is constant in anodic compartment with the working of cell
- d) $E_{\text{cell}} = E^\circ_{\text{Ag}^+/\text{Ag}} - E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}^+} + \frac{0.059}{1} \log \frac{1}{[\text{Cl}^-]}_{\text{anodic}}$
17. The cell emf depends on
- a) Temperature b) Size of cell c) Cathode electrolyte d) Anode electrolyte
18. Which of the following statements is/are correct ?
- a) ΔG° depends on cell reaction
- b) E°_{cell} depends upon making up of the cell
- c) Both ΔG° and E° are path functions
- d) E° is path function but ΔG° is a state function.
19. Which is/are concentration cells?
- a) $\text{Pt} : \text{H}_2/\text{HCl} // \text{HCl}/\text{H}_2 : \text{Pt}$
- b) $\text{Pt} : \text{H}_2 / \text{HCl} // \text{HCl}/\text{H}_2 : \text{Pt}$
 $(P^{\text{H}}=4) \quad P^{\text{H}}=2$
- c) $\text{Ag} / \text{AgCl} // \text{AgBr} / \text{Ag}$
 $(1\text{M}) \quad 1\text{M}$
- d) $\text{Ag} / \text{AgCl, KCl} // \text{AgNO}_3 / \text{Ag}$
 $(\text{s}) \quad 1\text{M}$

Linked Comprehension Type Questions

Passage-I :

The E° values for the changes given below are measured against NHE at 27°C

$\text{Cu}^{2+} + e \rightarrow \text{Cu}^+, E^\circ = +0.15 \text{ V}; \text{Cu}^+ + e \rightarrow \text{Cu}, E^\circ = +0.50 \text{ V}; \text{Zn}^{2+} + 2e \rightarrow \text{Zn}, E^\circ = -0.76 \text{ V}.$

The temperature coefficient of emf a cell designed as $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$ is $-1.4 \times 10^{-4} \text{ V per degree}.$

For a cell reaction in equilibrium $\Delta G = 0$ and $\Delta G^\circ = -2.303 RT \log K_c$. The heat of reaction and entropy change during the reaction are related by $\Delta G = \Delta H - T\Delta S$.

20. The E° for the reaction $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$ is
- a) +0.35 V b) -0.35 V c) +0.65 V d) -0.65 V
21. The decrease in free energy during the cell reaction in $\text{Zn} | \text{Zn}^{2+}(1\text{M}) || \text{Cu}^{2+}(1\text{M}) | \text{Cu}$, when its changes to 1M (Zn^{2+}) and 0.1M (Cu^{2+}) is
- a) $2.037 \times 10^5 \text{ J}$ b) $2.116 \times 10^5 \text{ J}$ c) $2.037 \times 10^6 \text{ J}$ d) $2.116 \times 10^6 \text{ J}$
22. The equilibrium constant for the reaction : $\text{Zn} + \text{Cu}^{2+} \rightleftharpoons \text{Zn}^{2+} + \text{Cu}$ is
- a) 6.0×10^{36} b) 1.94×10^{37} c) 1.47×10^{38} d) 7.0×10^{30}

Passage-II :

The driving force ΔG , diminishes to zero on the way to equilibrium, just as in any other spontaneous process. Both ΔG and the corresponding cell potential ($E = -\Delta G / nF$) are zero when the redox reaction comes to equilibrium. The Nernst equation for the redox process of the cell may be given as : $E = E^\circ - (0.059 / n) \log Q$. $E^\circ = (-\Delta G^\circ / nF)$; $E^\circ = (-0.059 / n) \log K$

23. On the basis of information available for the reaction : $\frac{4}{3} \text{Al} + \text{O}_2 \rightarrow \frac{2}{3} \text{Al}_2\text{O}_3$; $\Delta G = -827 \text{ kJ/mol}$ of O_2 , the minimum emf required to carry out an electrolysis of Al_2O_3 is :
- a) 2.14 V b) 4.28 V c) 6.42 V d) 8.56 V

24. If E^0 for the electro chemical cell, $\text{Zn(s)}|\text{Zn}^{2+}(1\text{M})\text{aq}||\text{Cu}^{2+}(1\text{M})\text{aq}|\text{Cu(s)}$ is 1.1V.

The $\frac{[\text{Cu}^{2+}]}{[\text{Zn}^{2+}]}$ for the cell reaction: $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq.}) \rightleftharpoons \text{Cu(s)} + \text{Zn}^{2+}(\text{aq.})$ will be :

- a) 10^{37} b) 10^{-37} c) 10^{-39} d) 10^{39}

Matrix Matching Type Questions

25. The standard reduction potential data at 25°C is given below:

$E^0(\text{Fe}^{3+}, \text{Fe}^{2+}) = +0.77\text{V}$; $E^0(\text{Fe}^{2+}, \text{Fe}) = -0.44\text{V}$; $E^0(\text{Cu}^{2+}, \text{Cu}) = +0.34\text{V}$; $E^0(\text{Cu}^+, \text{Cu}) = +0.52\text{V}$;

$E^0(\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}) = +1.23\text{V}$; $E^0(\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-) = +0.40\text{V}$;

$E^0(\text{Cr}^{3+}, \text{Cr}) = -0.74\text{V}$; $E^0(\text{Cr}^{2+}, \text{Cr}) = -0.91\text{V}$;

Match E^0 of the redox pair in Column-I with the values given in List-II and select the correct answer using the code given below the lists.

Column-I

- A) $E^0(\text{Fe}^{2+}, \text{Fe})$
 B) $E^0(4\text{H}_2\text{O} \rightleftharpoons 4\text{H}^+ + 4\text{OH}^-)$
 C) $E^0(\text{Cu}^{2+} + \text{Cu} \rightarrow 2\text{Cu}^+)$
 D) $E^0(\text{Cr}^{3+}, \text{Cr}^{2+})$

Column-II

- p) -0.18V
 q) -0.4V
 r) -0.04V
 s) -0.83V

Integer Type Questions

26. $\text{I}_2(\text{s})|\text{I}^-(0.1\text{M})$ half cell is connected to a $\text{H}^+(\text{aq})|\text{H}_2(1\text{bar})|\text{Pt}$ half cell and e.m.f. is found to be 0.7714 V. If $E^0_{\text{I}_2/\text{I}^-} = 0.535\text{V}$, find the pH of $\text{H}^+|\text{H}_2$ half-cell.

27. The E^0 at 25°C for the following reaction is 0.22 V $\text{H}_2(\text{g}) + 2\text{AgCl}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq})$. If the equilibrium constant at 25°C is 2.8×10^x ; x is

28. Given the following standard electrode potential, $\text{PbBr}_2(\text{s}) + 2\text{e}^- \rightarrow \text{Pb(s)} + 2\text{Br}^-(\text{aq})$; $E^0 = -0.248\text{V}$
 $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb(s)}$; $E^0 = -0.126\text{V}$. If the K_{sp} for PbBr_2 is 7.4×10^{-3} ; x is

29. If the equilibrium constant (K_c) for the reaction, $\text{Zn}^{2+}(\text{aq}) + 4\text{OH}^-(\text{aq}) \rightleftharpoons \text{Zn(OH)}_4^{2-}(\text{aq})$ is 10^{10x} ; then x is ($\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)}$; $E^0 = -0.76\text{V}$, $\text{Zn(OH)}_4^{2-}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)} + 4\text{OH}^-(\text{aq})$; $E^0 = -1.36\text{V}$)

30. Consider $\text{H}_2(\text{g})$, $\text{Pt}/\text{H}^+(\text{aq})$; $E = 0.1\text{V}$; the pH of the solution is :

EXERCISE-IV

(Electro Chemical Series & Batteries)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The metal that cannot displace hydrogen from dilute hydrochloric acid is
 1) aluminium 2) iron 3) copper 4) zinc
- A standard hydrogen electrode has zero electrode potential because
 1) Hydrogen is easiest to oxidise
 2) This electrode potential is assumed to be zero
 3) Hydrogen atom has only one electron
 4) Hydrogen is the lightest element

3. The correct statement among the following, about electrochemical series (S.R.P. increasing order) is
 1) the metals occupying top positions in the series do not liberate hydrogen with dilute acids
 2) the substances which are stronger reducing agents and stronger oxidising agents are placed below & top respectively
 3) a metal higher in the series will displace the metal from its solution which is lower in the series
 4) various electrodes are arranged in a series in the descending order of their potentials
4. The following are some statements about normal hydrogen electrode
 1) when a 'Zn' electrode is in combination of NHE, Zn electrode acts as cathode
 2) when a 'Cu' electrode is in combination with NHE, Cu electrode is the anode
 3) when a 'Ag' electrode is in combination with NHE, Ag electrode is the anode
 4) all are incorrect
5. Which defines the standard reduction electrode potential of Zn^{2+} ions?
 1) $\text{Zn}^{2+}_{(\text{IM})} + 2\text{e}^- \rightarrow \text{Zn}_{(\text{s})}$ 2) $\text{Zn}_{(\text{g})} \rightarrow \text{Zn}^{2+}_{(\text{IM})} + 2\text{e}^-$ 3) $\text{Zn}^{2+}_{(\text{IM})} \rightarrow \text{Zn}_{(\text{s})} + 2\text{e}^-$ 4) $\text{Zn}^{2+}_{(\text{g,IM})} \rightarrow \text{Zn}_{(\text{s})} - 2\text{e}^-$
6. (A) : Lithium has less electrode potential than cesium
 (R) : Hydration energy of lithium ion is high.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
7. The reference electrode is made by using
 1) ZnCl_2 2) CuSO_4 3) HgCl_2 4) Hg_2Cl_2
8. The more electro positive element has
 1) positive reduction potential 2) negative reduction potential
 3) tendency to gain electrons 4) negative oxidation potential
9. Stronger the oxidising agent greater is the
 1) Oxidation potential 2) Reduction potential 3) Redox potential 4) Hydration potential
10. Standard reduction electrode potential of three metals A, B and C are respectively +0.05 V, -3.0 and -1.2 V. The reducing powers of
 1) $\text{B} > \text{C} > \text{A}$ 2) $\text{A} > \text{B} > \text{C}$ 3) $\text{C} > \text{B} > \text{A}$ 4) $\text{A} > \text{C} > \text{B}$
11. Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because
 1) Zn acts as OA when reacts with HNO_3 2) HNO_3 is weaker acid than H_2SO_4 and HCl
 3) In electrochemical series Zn is above H_2 4) NO_3^- is reduced in preference to H_3O^+
12. When Zn piece is kept in CuSO_4 solution, copper gets precipitated because : (Z = Atomic number)
 1) $\text{SRP} = \text{Zn} > \text{Cu}$ 2) $\text{SRP} = \text{Zn} < \text{Cu}$ 3) $\text{Z} = \text{Zn} > \text{Cu}$ 4) $\text{Z} = \text{Zn} < \text{Cu}$
13. For $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$, standard reduction potential = +0.54 volt. For $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$, standard oxidation potential = -1.09 volt. For $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$, standard oxidation potential = +0.44 volt. Which of the following reactions is non-spontaneous?
 1) $\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$ 2) $\text{Fe} + \text{Br}_2 \rightarrow \text{Fe}^{2+} + 2\text{Br}^-$ 3) $\text{Fe} + \text{I}_2 \rightarrow \text{Fe}^{2+} + 2\text{I}^-$ 4) $\text{I}_2 + 2\text{Br}^- \rightarrow 2\text{I}^- + \text{Br}_2$
14. Be is placed above magnesium in the II group. When Be dust is added to MgCl_2 solution, it
 1) Has no effect 2) Precipitates Mg metal
 3) Precipitates MgO 4) Leads to dissolution of Be metal

15. When an electric cell is charged, then
 1) voltage of cell increases
 2) electrolyte of cell dilutes
 3) resistance of cell increases
 4) None of these
16. When lead storage battery is discharged
 1) SO_2 is evolved
 2) PbSO_4 is consumed
 3) lead is formed
 4) H_2SO_4 is consumed
17. When lead storage battery is charged
 1) PbO_2 dissolves
 2) Pb rod coated by PbSO_4
 3) H_2SO_4 is regenerated
 4) The amount of acid decreases
18. During the charging of lead storage battery, the reaction occurring at cathode is represented by
 1) $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$
 2) $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$
 3) $\text{Pb}^{2+} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4$
 4) $\text{PbSO}_4 + 2\text{H}_2\text{O} \rightarrow \text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^-$
19. With respect to fuel cell prepared from H_2 and O_2 gases, the false statement is
 1) It is free from pollution
 2) This is more efficient than conventional method of generating electricity
 3) The reaction occurring at anode is $\text{O}_{2(\text{g})} + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
 4) These take little time to go into operation.
20. Corrosion is basically a/an
 1) altered reaction in presence of H_2O
 2) electrochemical phenomenon
 3) interaction
 4) union between two light metals and a heavy metal
21. In which of the following will the corrosion of iron be most rapid ?
 1) In pure water
 2) In pure oxygen
 3) In air and moisture
 4) In air and saline water
22. Which of the following metals acts as a sacrificial anode for iron articles ?
 1) Cu
 2) Zn
 3) Ag
 4) Sn
23. The composition of rust is
 1) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
 2) $\text{Fe}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
 3) $\text{Fe}_2\text{O}_3 \cdot 6\text{H}_2\text{O}$
 4) Fe_2O_3
24. Zinc is used to protect corrosion of iron because
 1) $E_{\text{ox}} = \text{Zn} < E_{\text{ox}}$ of Fe
 2) $E_{\text{red}} = \text{Zn} < E_{\text{red}}$ of Fe
 3) cost = Zn < Fe
 4) Zn is most abundant

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The blue colour of aq. CuSO_4 solution, disappears, when 'x' is added, then $x = ?$
 a) Ag
 b) Mg
 c) Au
 d) Hg
2. An aqueous solution containing one mole per litre of each dil H_2SO_4 , AgNO_3 , $\text{Hg}(\text{NO}_3)_2$, $\text{Au}(\text{NO}_3)_3$ is being electrolysed using inert electrodes, then the sequence of deposition of elements at cathode will be
 a) $\text{Ag} > \text{Hg} > \text{H}_2 > \text{Au}$
 b) $\text{H}_2 > \text{Au} > \text{Hg} > \text{Ag}$
 c) $\text{Au} > \text{Ag} > \text{Hg} > \text{H}_2$
 d) $\text{H}_2 > \text{Hg} > \text{Ag} > \text{Au}$
3. Given $E_{\text{Ag}^+/\text{Ag}}^0 = 0.80 \text{ V}$, $E_{\text{Mg}^{2+}/\text{Mg}}^0 = -2.37 \text{ V}$, $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34 \text{ V}$, $E_{\text{Hg}^{2+}/\text{Hg}}^0 = 0.79 \text{ V}$.
 Which of the following statements is/are correct ?
 a) AgNO_3 can be stored in copper vessel
 b) $\text{Cu}(\text{NO}_3)_2$ can be stored in magnesium vessel
 c) CuCl_2 can be stored in silver vessel
 d) HgCl_2 can be stored in copper vessel

4. The half cell reactions for the corrosion are, $2\text{H}^+ + \frac{1}{2}\text{O}_2 + 2\text{e}^- \rightarrow \text{H}_2\text{O}$; $E^0 = 1.23\text{V}$ and $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe(s)}$; $E^0 = -0.44\text{V}$. Find the ΔG^0 (in kJ) for the overall reaction
 a) -76 K b) -322 kJ c) -161 kJ d) -152 kJ
5. The E^0 at 25°C for the following reaction at the indicated concentrations is 1.50 V. Calculate the ΔG in kJ at 25°C : $\text{Cr(s)} + 3\text{Ag}^+(\text{aq}, 0.1\text{M}) \rightarrow \text{Ag(s)} + \text{Cr}^{3+}(\text{aq}, 0.1\text{M})$
 a) -140.94 b) -295 c) -212 d) -422.83 kJ
6. A fuel cell develops an electrical potential from the combustion of butane at 1 bar and 298 K
 $\text{C}_4\text{H}_{10(\text{g})} + 6.5\text{O}_{2(\text{g})} \rightarrow 4\text{CO}_{2(\text{g})} + 5\text{H}_2\text{O}_{(\text{l})}$; $\Delta G^0 = -2746\text{kJ/mol}$
 a) 4.74 V b) 0.547 V c) 4.37 V d) 1.07 V
7. For a cell reaction $2\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{H}_2\text{O}_{(\text{l})}$, $\Delta S_{298}^0 = -0.32\text{kJ/K}$. What is the value $\Delta_f H_{298}^0(\text{H}_2\text{O}, \ell)$?
 Given : $\text{O}_{2(\text{g})} + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}_{(\text{l})}$; $E^0 = 1.23\text{V}$
 a) -285.07 kJ/mol b) -570.14 kJ/mol c) 285.07 kJ/mol d) None of these

More than One correct answer Type Questions

8. Based on the data given below, the correct order of reduction power is : $\text{Fe}^{3+}_{(\text{aq})} + \text{e}^- \rightarrow \text{Fe}^{2+}_{(\text{aq})}$, $E^0 = +0.77\text{V}$; $\text{Al}^{3+}_{(\text{aq})} + 3\text{e}^- \rightarrow \text{Al}_{(\text{s})}$, $E^0 = -1.66\text{V}$; $\text{Br}_{2(\text{aq})} + 2\text{e}^- \rightarrow 2\text{Br}^{-}_{(\text{aq})}$; $E^0 = 1.08\text{V}$
 a) $\text{Br}^- < \text{Fe}^{2+}$ b) $\text{Fe}^{2+} < \text{Al}$ c) $\text{Al} < \text{Br}^- < \text{Fe}^{2+}$ d) $\text{Al} < \text{Fe}^{2+} < \text{Br}^-$
9. If the half cell reaction $\text{A} + \text{e}^- \rightarrow \text{A}^-$ has a larger negative potential, it follows that :
 a) A is not readily reduced b) A is readily oxidised
 c) A^- is readily reduced d) A^- is readily oxidised
10. Which of the following facts are true?
 a) If $E^0(\text{M}^{n+}/\text{M})$ is negative, H^+ will be reduced to H_2 by the metal M.
 b) If $E^0(\text{M}^{n+}/\text{M})$ is positive, M^{n+} will be reduced to M by H_2
 c) In a cell, M^{n+}/M assembly is attached to hydrogen - half cell. To produce spontaneous cell reaction, metal M will act as negative electrode if the potential M^{n+}/M is negative. It will serve as positive electrode, if M^{n+}/M has a positive cell potential.
 d) Compounds of active metals (Zn, Na, Mg) are reducible by H_2 where as those of noble metals (Cu, Ag, Au) are not reducible.
11. Which of the following is/are correct ?
 a) If the solutions pressure which is in contact with a metal rod is greater than osmotic pressure then such metal half cell behaves as anode.
 b) Generally the solutions pressure tends to drive the ions into the solution.
 c) If the solutions pressure which is in contact with a metal rod is lesser than the osmotic pressure the solution tries to acquire -ve charge with respect to the metal (or) such half cell behaves as cathode.
 d) Solution's pressure has no contribution in deciding the nature of the electrode.
12. Identify the true ones
 a) Daniel cell is a reversible cell b) Lead storage battery is a reversible cell
 c) $\text{Zn}/\text{H}_2\text{SO}_4/\text{Ag}$ cell is reversible
 d) Any reversible cell will receive current flow from the external source when connected if its emf is less than that of the source.

13. When a lead storage battery is discharged:

- a) Weight of both electrodes decreases
b) PbSO_4 formed at both electrodes
c) H_2SO_4 consumed
d) H_2O is formed

14. In a lead storage battery,

- a) increase the emf with increase in the % H_2SO_4
b) the emf decreases with increase in the % H_2SO_4
c) equivalent weight of H_2SO_4 is 98
d) equivalent weight of $\text{H}_2\text{SO}_4 = 49$

Linked Comprehension Type Questions

Passage-I :

Redox reactions play a pivotal role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decided which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° values (with respect to normal hydrogen electrode) Using this data : $\text{I}_2 + 2e^- \rightarrow 2\text{I}^-$, $E^\circ = 0.54$; $\text{Cl}_2 + 2e^- \rightarrow 2\text{Cl}^-$, $E^\circ = 1.36\text{V}$; $\text{Mn}^{3+} + e^- \rightarrow \text{Mn}^{2+}$, $E^\circ = 1.50$; $\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$, $E^\circ = 0.77\text{V}$; $\text{O}_2 + 4\text{H}^+ + 4e^- \rightarrow 2\text{H}_2\text{O}$, $E^\circ = 1.23$.

15. Among the following, identify the correct statement :

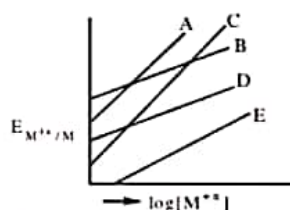
- a) Chloride ion is oxidised by O_2
b) Fe^{2+} is oxidised by iodine
c) Iodide ion is oxidised by chlorine
d) Mn^{2+} is oxidised by chlorine

16. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because :

- a) O_2 oxidises Mn^{2+} to Mn^{3+}
b) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
c) Fe^{3+} oxidises H_2O to O_2
d) Mn^{3+} oxidises H_2O to O_2

Passage-II :

Four metal electrodes show the variation in their reduction potential as shown.



17. Which is the most reactive metal ?

- a) A
b) B
c) E
d) D

18. Which statement is true ?

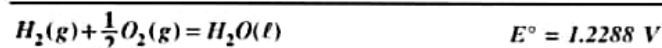
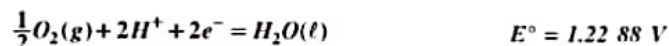
- a) B can reduce A^{n+} (1 M)
b) E cannot replace H^+ from acids
c) 'A-D' cell has maximum E° among the given
d) 'C' can reduce the ions of A, B and D

19. Which metal requires maximum voltage for deposition of one mole through electrolysis process?

- a) A
b) B
c) C
d) D

Passage-III:

Hydrogen-oxygen fuel cell may have an acidic or alkaline electrolyte. The half-cell reactions are



20. If 560 mL of H_2 gas at STP is fed into and is consumed by the fuel cell in 10 minutes, then what is the current output of the fuel cell?
 a) 4A b) 8A c) 16A d) 12A
21. For a Hydrogen-Oxygen fuel cell if $\Delta H_f^\circ (H_2O, l) = -285 \text{ kJ/mole}$, then what will be its thermodynamic efficiency under standard conditions
 a) 0.91 b) 0.41 c) 0.63 d) 0.83
22. How much H_2O is produced in the cell if 2 Faradays of charge is derived from the fuel cells?
 a) 18 g b) 36 g c) 42 g d) 9 g

Matrix Matching Type Questions**23. Column-I**

- A) Non spontaneous reaction is made to occur
 B) Spontaneous redox reaction occurs
 C) No chemical reaction occurs
 D) Reaction can be reversed

Column-II

- p) Secondary voltaic cell
 q) Conductivity cell
 r) Dry cell
 s) Nelson cell

24. Column-I

- A) Primary cell
 B) Secondary cell
 C) Calomel electrode
 D) Hydrogen electrode

Column-II

- p) Lead storage cell
 q) Daniel cell
 r) Secondary reference cell
 s) Primary reference cell

KEY SHEET (LECTURE SHEET)**EXERCISE-I****LEVEL-I**

- 1) 4 2) 3 3) 3 4) 4 5) 1 6) 3 7) 4 8) 4
 9) 3 10) 2 11) 2 12) 1 13) 2 14) 1 15) 2 16) 3
 17) 3 18) 2 19) 3 20) 4 21) 4 22) 3 23) 3.54 24) 11.2
 25) 965 26) 10 27) 65 28) 4.5 29) 965

LEVEL-II

- 1) d 2) d 3) c 4) d 5) a 6) abcd 7) ac 8) bc
 9) ad 10) ac 11) bc 12) bc 13) d 14) b 15) b 16) d
 17) a 18) d 19) A-pq; B-pq; C-rq; D-ps
 20) A-r; B-s; C-p; D-q 21) 2 22) 4 23) 6 24) 8 25) 2
 26) 1 27) 3 28) 1

EXERCISE-II

LEVEL-I

- 1) 1 2) 3 3) 4 4) 2 5) 1 6) 4 7) 2 8) 4
 9) 1 10) 1 11) 3 12) 4 13) 4 14) 4 15) 3 16) 3
 17) 4 18) 4 19) 0.5 20) 220 21) 128 22) 130 23) 18.78
 24) 0.2 25) 391 26) 1.1 27) 1

LEVEL-II

- 1) c 2) c 3) b 4) a 5) b 6) ad 7) c 8) a
 9) a 10) b 11) a 12) c 13) b 14) b
 15) A-p; B-q; C-s; D-r 16) A-qs; B-pr; C-qs; D-pr 17) 6 18) 2
 19) 1 20) 8 21) 9 22) 5

EXERCISE-III

LEVEL-I

- 1) 3 2) 2 3) 3 4) 3 5) 1 6) 4 7) 3 8) 4
 9) 1 10) 3 11) 1 12) 1 13) 3 14) 4 15) 3 16) 4
 17) 4 18) 3 19) 1 20) 1 21) 2 22) 0.28 23) 0.01 24) 0.35
 25) 0.91 26) 0.80 27) 1.79 28) 0.52

LEVEL-II

- 1) c 2) c 3) a 4) a 5) b 6) c 7) d 8) d
 9) a 10) abc 11) ab 12) ad 13) bc 14) abc 15) acd 16) bcd
 17) acd 18) abd 19) abcd 20) a 21) a 22) a 23) a 24) b
 25) A-r; B-s; C-p; D-q 26) 3 27) 7 28) 5 29) 2 30) 2

EXERCISE-IV

LEVEL-I

- 1) 3 2) 2 3) 3 4) 4 5) 1 6) 1 7) 4 8) 2
 9) 2 10) 1 11) 4 12) 2 13) 4 14) 1 15) 1 16) 4
 17) 3 18) 4 19) 3 20) 2 21) 4 22) 2 23) 1 24) 2

LEVEL-II

- 1) b 2) c 3) c 4) b 5) d 6) d 7) a 8) ab
 9) ad 10) abc 11) abc 12) abd 13) acbd 14) ac 15) c 16) d
 17) c 18) d 19) c 20) b 21) d 22) a
 23) A-ps; B-pr; C-q; D-p 24) A-q; B-p; C-r; D-s